Amir Esmaeilpourmoghaddam

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EDUCATION

Expected 2028 **Ph.D. in Computer Science**
University of Illinois Chicago, Chicago, IL

2022 **B.Sc. in Computer Engineering**
K. N. Toosi University of Technology, Tehran, Iran
Thesis: Windshield Reflection Removal

EXPERIENCE

Fall 2023 – Present **Ph.D. Research Assistant**, University of Illinois Chicago
Research on RL preference alignment and bandits

Summer 2021 **Machine Learning Intern**, Technical University of Munich
Motion planning for autonomous vehicles using RL (Prof. Matthias Althoff)

2019 – 2020 **Teaching Assistant**
Fall 2020: Signal and Systems — Dr. Fatemeh Rezaei
Fall 2019: Hardware-Software Co-design — Dr. H. Roodaki

RESEARCH INTERESTS

- Reinforcement Learning from Human Feedback (RLHF) & Preference Alignment
- Bandit Algorithms & Online Learning
- Computer Vision & Image Processing
- Natural Language Processing

SELECTED COURSE PROJECTS

Spring 2025 **Knowledge Distillation Comparison** — UIC CS 594
Implemented logit imitation, feature imitation, and feature imitation with orthogonal projection.

Fall 2024 **iVAE: Nonlinear ICA** — UIC CS 520
Implemented Identifiable VAE and compared against vanilla VAE on ICA benchmarks.

Fall 2023 **SNN Classifier & Transformer Language Model** — UIC CS 412
Built a spiking neural network classifier on MNIST and an attention-based transformer from scratch for citation classification.

Personal **Edge Motion Detection via Discrete MRF & Belief Propagation**
Optical flow estimation using Markov Random Fields and loopy belief propagation.

Spring 2021 **BEV Perspective of Soccer Match** — KNTU Computer Vision
Produced bird's-eye-view perspective from a soccer live stream using homography estimation (PyTorch, OpenCV).

Spring 2020 **Stanford CS231n Projects**
LSTM image captioning on COCO; GAN variants (standard, DC-GAN, LS-GAN) for image generation on MNIST.

SELECTED COURSEWORK

Spring 2025	UIC CS 512 — Advanced Machine Learning (Prof. Xinhua Zhang)
Fall 2025	UIC ECE 508 — Convex Optimization (Prof. Shuo Han) — A
Spring 2025	UIC CS 594 — Energy-Efficient Deep Learning (Prof. Yan Yan) — A
Fall 2024	UIC CS 520 — Causal Inference and Learning (Prof. Elena Zheleva)
Fall 2023	UIC CS 412 — Introduction to Machine Learning (Prof. Pedram Rooshenas) — A
2020	MIT 18.06 — Linear Algebra (Prof. Gilbert Strang)
2020	MIT RES.6-012 — Introduction to Probability (Prof. John Tsitsiklis)
2020	Stanford CS231n — Convolutional Neural Networks for Visual Recognition (Prof. Fei-Fei Li)
2020	Coursera — Deep Learning Specialization (Prof. Andrew Ng)
2019	Georgia Tech — Introduction to Computer Vision (Prof. Aaron Bobick)

TECHNICAL SKILLS

Languages: Python, Java, C, C++, MATLAB, SQL, Assembly

Frameworks: PyTorch, TensorFlow, OpenCV, scikit-learn

AWARDS

2017	Ranked 180th among 30,000+ applicants in the nationwide university entrance exam (Iran)